

Tutorial 2: Student Attendance Portal

Course: 24CSC42 Full Stack Development

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1. Scenario

The objective of this tutorial is to build a full-stack 'Student Attendance Portal' using React.js for the frontend. The application communicates with a backend API (using Axios) to perform CRUD operations on student attendance records. Key features include fetching records (GET), updating attendance status (PUT) via a styled dropdown, deleting incorrect entries (DELETE), and a seeding mechanism to load sample data. The UI includes a real-time statistics bar (Total, Present, Absent, OD) and toast notifications for user feedback.

2. VS Code Project Structure

```
student-attendance-portal/
├── src/
│   ├── components/
│   │   ├── AttendanceTable.js    # Main data grid with status controls
│   │   └── Toast.js              # Notification component
│   ├── App.js                    # State management and API integration
│   ├── index.css                 # Navy, Gold, and Status-based styling
│   └── index.js                  # Entry point
├── package.json
└── README.md
```

3. Source Code

App.js

```
import React, { useState, useEffect, useCallback } from 'react';
import axios from 'axios';
import AttendanceTable from '../components/AttendanceTable';
import Toast from '../components/Toast';

const API = '/api/attendance';

function App() {
  const [records, setRecords] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState('');
  const [toast, setToast] = useState(null);

  const showToast = (msg, type = 'success') => {
    setToast({ msg, type });
    setTimeout(() => setToast(null), 3000);
  };

  const fetchAttendance = useCallback(async () => {
    setLoading(true);
    try {
      const { data } = await axios.get(API);
      setRecords(data);
    } catch (err) {
      setError('Failed to fetch attendance.');
```

```

newStatus });

    setRecords((prev) => prev.map((r) => (r._id === id ? data : r)));
    showToast(`Status updated to "${newStatus}"`, 'success');
  } catch (err) { showToast('Failed to update status.', 'error'); }
};

const handleDelete = async (id, name) => {
  if (!window.confirm(`Delete record for ${name}?`)) return;
  try {
    await axios.delete(`${API}/${id}`);
    setRecords((prev) => prev.filter((r) => r._id !== id));
    showToast(`Record for "${name}" deleted.`, 'info');
  } catch (err) { showToast('Failed to delete.', 'error'); }
};

// ... (Stats and JSX)
}

```

AttendanceTable.js

```

function AttendanceTable({ records, loading, error, onStatusChange,
onDelete }) {
  if (loading) return <div className="loading-wrap">Fetching
records...</div>;

  if (error) return <div className="error-wrap">⚠ {error}</div>;

  return (
    <div className="table-card">
      <div className="table-header-row">
        <span className="th">Roll No.</span>
        <span className="th">Student Name</span>
        <span className="th">Status</span>
        <span className="th">Action</span>
      </div>
      {records.map((rec) => (
        <div className="att-row" key={rec._id}>
          <span className="cell-roll">{rec.rollNo}</span>
          <span className="cell-name">{rec.name}</span>
          <span>
            <select

```

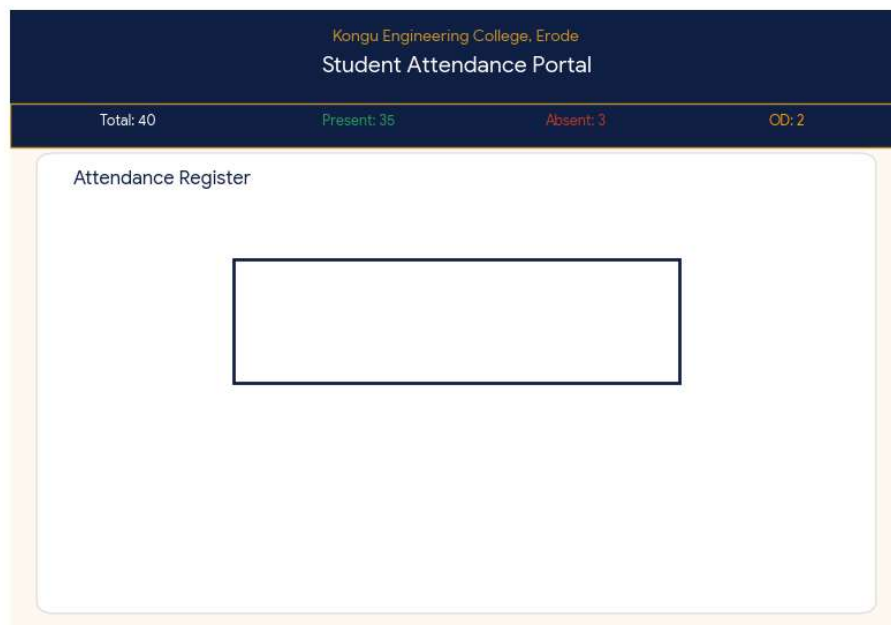
```

        className={`status-select ${statusClass(rec.status)}`}
        value={rec.status}
        onChange={(e) => onStatusChange(rec._id, e.target.value)}
      >
        <option value="Present">Present</option>
        <option value="Absent">Absent</option>
        <option value="On Duty">On Duty</option>
      </select>
    </span>
    <button className="del-btn" onClick={() => onDelete(rec._id,
rec.name)}>X</button>
  </div>
  )}
</div>
);
}

```

4. Output Screenshots

4. Delete Confirmation



3. Status Update Success

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Student Attendance Portal

Total: 40Present: 35Absent: 3OD: 2

Attendance Register

[Register View]

2. Attendance Register View

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Student Attendance Portal

Total: 40Present: 35Absent: 3OD: 2

Attendance Register

22CSR001	Aditya R	Present	×
22CSR002	Bhavani S	Absent	×

1. Loading State

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Student Attendance Portal

Total: 40

Present: 35

Absent: 3

OD: 2

Attendance Register

Fetching attendance records from server...